EXP NO-5
DATE-

IMPLEMENT YOUR OWN PING PROGRAM

AIM: To implement a simple Ping program using socket programming to check whether a host is reachable and to calculate the Round Trip time (RTT).

PROCEDURE:

1) Import the required socket and time modules.
2) Get the hostname or IP address of the destination host.
3) Create a socket connection with the destination host.
4) Record the time before sending the request.
5) Send a request to the destination server.
6) Wait for the response from the server.
7) Record the time when the response is received.
8) Calculate the Round Trip Time (RTT).
9) Display the destination address, response time, and packet status.
10) Repeat the process for a few packets and display the results.

ALGORITHM:

1) Start
2) Get the destination hostname / IP address.
3) Resolve the hostname into an IP address.
4) Create a socket.
5) Record the start time.
6) Send a request to the destination.
7) Receive the response.
8) Calculate RTT: End Time - Start Time
9) Display the RTT and status.
10) Repeat the required number of packets.
11) Stop.

PROGRAM:

```
import socket
import time
host = input("Enter hostname or IP address: ")
try:
    ip = socket.gethostbyname(host)
    print("Pinging", host, "[", ip, "]")
    for i in range(4):
        start = time.time()
        try:
            A = socket.create_connection((host, 80),
                                         timeout=2)
            s.close()
            end = time.time()
            rtt = (end - start) * 1000
            print("Reply from", ip, ": time =", round(
                rtt, "ms")
        except: print("Request timed out")
        time.sleep(1)
except socket.gaierror:
    print("Host not found")
```

SAMPLE OUTPUT:

```
Enter hostname or IP address : google.com
Pinging google.com [142.250.195.14]
Reply from 142.250.195.14 : time = 24.31 ms
Reply from 142.250.195.14 : time = 22.87 ms
Reply from 142.250.195.14 : time = 23.45 ms
Reply from 142.250.195.14 : time = 25.12 ms
```

RESULT: Thus, the Ping program was successfully implemented using socket programming to check host reachability and calculate the Round Trip time (RTT).